27) $i \hookrightarrow \$t_1$
result $\hookrightarrow \$s_2$
& MemArray $\hookrightarrow \$s_0$

```
for (int i=0; i<100; i+=1)
{
  result = result + MemArray[i];
}
```

```
addi $t1, $0, 0      (i = 0)
Loop: lw $s1, 0($s0)  ($s1 = MemArray[0])
add $s2, $s2, $s1
      (result = result + MemArray[0])
addi $s0, $s0, 4
      ($s0 = &MemArray + 1)
addi $t1, $t1, 1
      (i = i+1)
slti $t2, $t1, 100
      ($t2 = $t1 < 100)
bne $t2, $0, Loop
```

HW3: 1) 5ED4
+ 07A4
5730

4 - 4 = 0
D - A = 13 - 10 = 3
E - 7 = 14 - 7 = 7
5 - 0 = 5

unsigned 8 bit 6) 185
+ 122

1011 1001 → neg
0111 1010 → in
0011 1111 → p
⇒ neither

g) 1001 0111 (151) → −105 because of 8-bit signed
1101 0110 (214) → + −42 negative
0110 1101 → 109 → overflow
⇒ Overflow so based on saturating arithmetic
it should be: −128 or 1000 0000

1000 0000
0111 1111

128 01000000
1011 1110
11000000

11) 151 : 1001 0111 unsigne 8-bit
214 : +1101 0110
0110 1101 ⇒ overflow
So based on saturating arithmetic ⇒ answer stopped at 1111 1111.
or 255

12) 8 bit: 62 → 00111110 (multicand)
12 → 00001100 (multiplier)

			Mult	MC	Product
62	0	0	00001100	0000 0000 0011 1110	0000 0000 0000 0000
124		1	00000110	0000 0000 0111 1100	0000 0000 0000 0000
248		2	00000011	0000 0000 1111 1000	0000 0000 0000 0000
496	248	3	00000001	0000 0001 1111 0000	0000 0000 1111 1000
	744	4	00000000	0000 0011 1110 0000	0000 0010 1111 1000
		5	00000000	0000 0111 1100 0000	0000 0010 1111 1000
		6	00000000	0000 1111 1000 0000	0000 0010 1111 1000
		7	00000000	0001 1111 0000 0000	0000 0010 1111 1000
		8	00000000	0011 1110 0000 0000	0000 0010 1111 1000

(octal) $62_8 = 110\,010_2 \Rightarrow$ 8 bit : $0011\,0010_2$
$12_8 = 001\,010_2 \Rightarrow$: $0000\,1010_2$

			Mult	MC	Product
50	0	0	0000 1010	0000 0000 0011 0010	0000 0000 0000 0000
100	0	1	0000 0101	0000 0000 0110 0100	0000 0000 0000 0000
200	100	2	0000 0010	0000 0000 1100 1000	0000 0000 0110 0100
400	100	3	0000 0001	0000 0001 1001 0000	0000 0000 0110 0100
000	500	4	0000 0000	0000 0011 0010 0000	0000 0001 1110 0100
		5	0000 0000	0000 0110 0100 0000	0000 0001 1110 0100
		6	0000 0000	0000 1100 1000 0000	0000 0001 1110 0100
		7	0000 0000	0001 1001 0000 0000	0000 0001 1110 0100
		8	0000 0000	0011 0010 0000 0000	0000 0001 1110 0100

11 1100 → 010001

60/17 → 1. 1st version of division, 6 bit => 7 ops

18)

			Q	DIV	Rem
1088	60	0	000 000	010001 000.000	000 000 111100
544	60	1	000000	001000 100 000	000 000 111100
272	60	2	000 000	000100 010000	000 000 111100
136	60	3	000 000	000010 001000	000 000 111100
68	60	4	000 000	000001 000100	000 000 111100
34	60	5	000000	000000 100010	000 000 111100
17	26	6	000001	000 000 010001	000 000 011010
9	7		000011	000 000 001000	000 000 001001

19) optimized, 6 bit, 60/17

		Div	Rem
	0	010001	000 000 111100
30	1	010001	000 001 111000
17 60	2	010001	000011 110000
60 13	3	010001	000111 100000
24	4	010001	001111 000000
26	5	010001	001101 000001
	6	010001	001001 000011

22) 0000 1100 0000 0000 0000 0000 0000 0000

sign: 0 ⇒ pos

exp: 0001 1000 ⇒ 24 | Bias: 127

exp − bias = −103

⇒ $1.000\ldots0 \cdot 2^{-103} = 2^{-103}$

Double

S	Exp (11)	Frac (52)	GRS
0	10000000100	1111101...0	

24) 63.25

pos ⇒ sign = 0

int = $63_{10} = 111111_2$

frac: .01

$\Rightarrow 11111.01 \cdot 2^0$

$= 1.111101 \cdot 2^5$

⇒ exp − 1023 = 5

⇒ exp = 1028 = 10000000100

23) exp − 127 = 5

exp = 132 = 1000 0100

⇒ 0 1000 0100 1111101...0

sign | 8 bits exp | 23 bits frac

25) 63.25 ; 7 bits of exponent

69 − 48 = 15

Sign: 0

int: $3F.4_{16} \cdot 16^0$

Bias: $2^{7-1} = 64$

$\Rightarrow 3.F4_{16} \cdot 16^1 \Rightarrow 0.3F4_{16} \cdot 16^2$

exp − 64 = 2 ⇒ exp = 66 → 1000010_2

frac: 3F4: 0011 1111 0100

⇒ 0 1000010 0011 1110 1...0

1 bit | 7 bit | 24 bits

26) $-1.5625 \cdot 10^{-1} = -0.15625 = -0.625 \cdot 2^{-2}$

Since exp is −2

consider +2: 0000 0000 0010

⇒ −2: 1111 1111 1110

0.625 for fraction: 0.1010 0000 0000 0000 0000 000

⇒ −0.625: 1 10110 0000 0000 0000 0000

→ 1111 1111 1110 1011 0000 0000 0000 0000 0000 0000

Exp (12 bit) — 24 bit fraction

Range for exp: −2048 → 2047 compared to 8 bit and 11 bits of IEEE 754 (very large)

Accuracy: 24 bit fraction = $24 \cdot \log_{10}(2) \approx 7$ digits

≈ sigle precision

= $\frac{1}{2}$ of double precision (15 digits)

27) 16-bit half precision.

sign: 1 bit

exp: 5 bits → bias = 15 (lol, excess-16 ⇒ bias = 16)

frac: 10 bits

$-1.5625 \cdot 10^{-1} \approx -0.15625$

⇒ sign = 1 ; fract: .00101

⇒ $-0.00101 \cdot 2^0$

⇒ $-1.01... \cdot 2^{-3}$ ⇒ exp − bias = −3

→ 1 01101 00101 ⇒ exp = 15 − 3 = 13

00000 ⇒ 1101

Range of exp: −15 to 14 vs −126 to 127

Range: 2^{-15} to 2^{15} vs 2^{-126} to 2^{128}

accuracy: 10 bits ⇒ $10 \cdot \log_{10}(2)$ ≈ 7 digits

fract ≈ 3 digits

⇝ frac = 10 bits

29) 16-bit A, B, 7 bit for G, R, S.

$2.6125 \cdot 10^1 = 26.125$

$+ 4.150390625 \cdot 10^{-1} = 0,4150390625$

26.125 ⇒ sign = 0

int = 11010.001 $\cdot 2^0$ | ⇒ fract: .01101010010

fract ⇒ $1.1010001 \cdot 2^4$ | ⇒ 0.01101|0010 $\cdot 2^0$

⇒ $1.10100|01000 \cdot 2^4$ | ⇒ 0.00000|11010|10 1 $\cdot 2^4$

↑ ↑ ↑ G R S

GRS

$1.10100\,01000 \mid 000 \cdot 2^4$

$+\ 0.00000\,11010 \mid 100 \cdot 2^4$

$1.10101\,00010 \mid 101 \cdot 2^4$

$1.10101\,00010 \mid 101$ N $\cdot 2^4 \rightarrow$ OF/UF ∅

$1.10101\,00011 \mid 000$ R $\cdot 2^4$

$1.10101\,00011 \mid 000$ N $\cdot 2^4 \rightarrow$ OF/UF ∅

$1.10101\,00011 \mid 000$ R $\cdot 2^4$

$\Rightarrow 1101010001.1_2 \cdot 2^{-6}$

$= 1699_{10} \cdot \frac{1}{64} \approx 26.5458\,75$

30) -8.0546875

16-bit int: 1000

fract: .0000111

$\Rightarrow -1000.0000111 \cdot 2^0$

$\Rightarrow -1.0000000|111 \cdot 2^3$

GRS

1000

$-1.7993164062\,5 \cdot 10^{-1}$

$= -0.1799316406\,25$

GRS

fract: $0.00101|1000|010 \cdot 2^0$

$\Rightarrow 1.0111|0000|101 \cdot 2^{-3}$

Sign $(-1) \times (-1) = 1$

exp: ~~(0+127)~~ $0+3=03$

GRS

$1.00000\,00111\ 000 \cdot 2^3$

$1.0111|0000|101\ 000 \cdot 2^{-3}$

$1.0000000111\ 000$

1000000110

1000000111

1000000011

10000

$1.0111\,0000\,1011|100 \cdot 2^0$

$1.0111\,0000\,1011 \cdot 2^0$

0.000000|00000.01.

=) 1483 : 1024 = 1.448242187

correct result: 1.449293131

diff by : 10.0725%

32) 16 bit → 10bit frac. → 5 bit exp → bias = 15

$3.984375 \cdot 10^{-1} = 0.3984375$

frac: 0.0110011 ⇒ $0.0110011 \cdot 2^{0}$

⇒ $1.10011 \cdot 2^{-2}$

$3.4375 \cdot 10^{-1} = 0.34375$

frac: .0101100000000000000

⇒ $0.01011000000000000 \cdot 2^{0}$

⇒ $1.01100000000000000 \cdot 2^{-2}$

⇒ $1.01100\ 00000 \cdot 2^{-2}$

$3.9874375 \cdot 10^{-1} + 3.4375 \cdot 10^{-1}$

⇒ $1.10011\ 00000\ 000 \cdot 2^{-2}$ (GRS)

$+\ 1.01100\ 00000\ 000 \cdot 2^{-2}$

$10.11111\ 00000\ 000 \cdot 2^{-2}$

OF/UF ← N ⇒ $1.01111\ 10000\ 000 \cdot 2^{-1}$

← R ⇒ $1.01111\ 10000\ 000 \cdot 2^{-1}$

N, R ⇒ $1.01111\ 10000 \cdot 2^{-1}$

$1.771 \cdot 10^{3} = 1771$ ⇒ int = 11011101011 $\cdot 2^{0}$

⇒ $1.10111\ 01011 \cdot 2^{10}$

N → $0.00000\ 00000\ 101 \cdot 2^{10}$ (GRS)

⇒ $1.10111\ 01011\ 000 \cdot 2^{10}$

$0.00000\ 00000\ 101 \cdot 2^{10}$

$1.10111\ 01011\ |\ 101 \cdot 2^{10}$

OF/UF ← N, R → $1.10111\ 01100\ |\ 000 \cdot 2^{10}$

N, R ⇒ $1101110111_{2} \cdot 2^{2}$

(OK) ⇒ $443 \cdot 2^{2} = \boxed{1772}$

10 + 15 ⇒ 25 ⇒ exp = 11001

16-bit FP ⇒ 0. 11001. 1011101100.

33) 16-bit → 1S, 5E, 10F

• $3.4375 \cdot 10^{-1} + 1.771 \cdot 10^{3}$

1. $3.4375 \cdot 10^{-1} = 0.34375$

⇒ fract: $0.01011 \cdot 2^{0}$

$\Rightarrow 1.011 \cdot 2^{-2}$

2. $1.771 \cdot 10^{3} = 1771$

→ int 11011|01011 $\cdot 2^{0}$

⇒ $1.10111\ 01011 \cdot 2^{10}$

$0.00000\ 00000\ 010 \cdot 2^{10}$

+ ⇒ $1.10111\ 01011\ \overset{G\ R\ S}{000} \cdot 2^{10}$

$0.00000\ 00000\ 010 \cdot 2^{10}$

$1.10111\ 01011\ 010 \cdot 2^{10}$

⇒ $1.10111\ 01011\ |\ 000$ N, R, N, R $\cdot 2^{10}$

N

• $3.984375 \cdot 10^{-1} = 0.3984375$

→ fract: .01100011 $\cdot 2^{0}$

⇒ $1.10011 \cdot 2^{-2}$

N→2^{10} ⇒ $0.00000\ 00000\ 011 \cdot 2^{10}$

⇒ Sum2 = $0.00000\ 00000\ |011 \cdot 2^{10}$

+ $1.10111\ 01011\ |000 \cdot 2^{10}$

$1.10111\ 01011\ |011 \cdot 2^{10}$

$1.10111\ 01011\ |000$ ← N, R, OF/UF = N, R, OF/UF $\cdot 2^{10}$

⇒ Sign = 0

exp = 10+15 = 25

= 11001

fract = .10111 01011

→ $0.11001.10111\ 01011$

decimal ⇒ $1101110101_{2} \cdot 2^{0}$

⇒ $1771 \cdot 1 =$ **1771**

34) ⇒ 1772 ≠ 1771 ⇒ NO

16-bit

35) • $3.41796875 \cdot 10^{-3} = 0.00341796875$ (1367187500)

⇒ fract: .0000000111

⇒ $0.0000000111 \cdot 2^{0}$

$1.1100000000 \cdot 2^{-9}$

• $6.34765625 \cdot 10^{-3} = 0.00634765625$ (5078124996 81249999936)

⇒ fract: .0000001101

⇒ $1.101 \cdot 2^{-8}$

sign = pos + pos = pos

$= 1.110\,000 \cdot 2^{-9}$ (GRS)

$\times$ $1.101\,000 \cdot 2^{-8}$

1.110 000

0.111 000

0.001 110

10.110 | 110 ↓ 2^{-17}

$= 1.011 | 0110 \cdot 2^{-16}$ N ⇒ (UF) ?

⇒ = 0

min(exp) = -14

Product = [0] ⇒ 0.00000.000000000000

36). $1.05625 \cdot 10^{2} = 105.625$

int: 1101001 ; fract: .101

⇒ $1101001.101 \cdot 2^{0}$

⇒ $1.101001101 \cdot 2^{6}$

first prod ⇒ $1.101001101 \cdot 2^{6}$

845

M64

$\times$ $1.1010000000 \cdot 2^{-8}$ (GRS)

1.101001101 000

0.110100110 100

0.001101001 101 ↓

10.101011101 | 001 2^{-2}

$1.010101110\,1 | 001 \cdot 2^{-1}$

$1.010101110\,1 | 000 \cdot 2^{-1}$

OF/UF

T

N

R

2nd Prod: $1.010101\,1101\,000 \cdot 2^{-1}$ (GRS)

$\times$ $0.1100000\,0000\,000 \cdot 2^{-9}$

$1.010101\,1101\,000$

$0.101010\,1110\,100$

$0.010101\,0111\,010$

$10.010110\,0001\,0|110 \cdot 2^{-10}$ OF/UF ∅

$1.0010\,1|0000\,1|011 \cdot N \cdot 2^{-9}$

$1.0010\,1000\,1 \cdot 2^{-9}$ R, N, R

⇒ sign = 0, exp = $-9+15 = 6 = 00110$

fract = 00101 10001

16b FP ⇒ 0 00110 00101 10001

⇒ Decimal = $100101\,10001_2 \cdot 2^{-19}$

$= 1201 \cdot \frac{1}{524288}$

≈ 0.002290725

3?) ⇒ (No)